

CASE STUDY-CO1-AT2

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COURSE NAME: REINFORCEMENT LEARNING

1. RL-Based Warehouse Robot (MDP)

MDP Components

- **States (S):** The state represents the current condition of the robot, including its location in the warehouse, battery level, obstacle positions, item location, and whether an item has been picked or delivered.
- **Actions (A):** The robot can move in four directions (Up, Down, Left, Right), pick an item, pack it, transport it, deliver it, or recharge the battery.
- **Environment:** The warehouse contains shelves, pathways, obstacles, charging stations, and delivery locations. The environment changes as tasks are completed.
- **Policy (π):** The policy determines the best action to take in each state to maximize the total reward.
- **Reward (R):** Rewards guide the robot toward efficient and safe task completion.

Reward Function

- +10 for successful delivery.
- +5 for correctly picking an item.
- -2 for unnecessary delay.
- -10 for hitting an obstacle.
- -1 for each extra movement.

Evaluation

The reward function balances **speed, accuracy, and safety**. A high reward for successful delivery encourages fast task completion, while collision penalties ensure safe navigation. If rewards are poorly designed, the robot may take unsafe shortcuts or waste time, reducing overall warehouse efficiency.

2. Movie Recommendation System using RL

State Representation

The state contains information about the user such as:

- Viewing history.
- Favorite genres.
- Recently watched movies.
- Ratings given by the user.
- Time of day and device used.

Action Space

The RL agent can:

- Recommend different movies.
- Recommend TV shows.
- Suggest new genres.
- Display personalized content lists.

Reward Function

- +5 if the movie is watched completely.
- +3 if the user gives a positive rating.
- +2 if the user watches another recommendation.
- -4 if the recommendation is skipped.

Exploration vs Exploitation

Exploration helps discover new user interests by recommending unfamiliar content, while exploitation recommends movies similar to previous preferences. A balance between both improves long-term user satisfaction.

Evaluation

Different reward signals directly affect engagement. Positive rewards encourage accurate recommendations, while negative rewards reduce poor suggestions. A balanced strategy increases viewing time and customer retention.

3. Autonomous Agricultural System

States

The system observes:

- Soil moisture level.
- Temperature.
- Humidity.
- Weather forecast.
- Crop growth stage.
- Nutrient level.

Actions

The agent can:

- Increase irrigation.
- Reduce irrigation.
- Apply fertilizer.
- Delay watering.
- Take no action.

Reward Function

- +10 for healthy crop growth.
- +5 for conserving water.
- -5 for overwatering.
- -5 for excessive fertilizer use.
- -8 for crop damage.

Delayed Rewards

Crop growth is observed only after several days or weeks. Therefore, the agent must learn that today's irrigation decisions influence future crop yield.

Suitable Learning Algorithm

Q-Learning is suitable because it learns the optimal policy without requiring knowledge of the environment model. It can improve decisions through repeated interactions.

Justification

Q-Learning effectively handles uncertain environments and delayed rewards, making it suitable for smart agriculture applications.

4. Cybersecurity Intrusion Detection

States

The system monitors:

- Network traffic.
- Login attempts.
- Packet transmission rate.
- CPU usage.
- Number of suspicious connections.
- Firewall status.

Environment

The environment consists of normal users and attackers generating different network activities. The system continuously observes these activities.

Actions

The RL agent can:

- Allow traffic.
- Block suspicious IP addresses.
- Generate alerts.
- Perform deep packet inspection.
- Isolate infected systems.

Reward Function

- +10 for correctly detecting attacks.
- +5 for blocking malicious traffic.
- -10 for allowing attacks.
- -5 for false alarms.

Evaluation

Cyber attacks occur infrequently, creating sparse rewards. Real-time detection also requires fast decision-making with minimal processing delay. Therefore, efficient learning and quick response are essential.

5. Ride-Sharing Driver Allocation

States

The system considers:

- Driver locations.
- Passenger requests.
- Traffic conditions.
- Weather.
- Time of day.
- Number of available vehicles.

Actions

The RL agent may:

- Assign a nearby driver.
- Increase fare.
- Reduce fare.
- Wait for additional drivers.
- Redirect drivers to high-demand areas.

Reward Function

- +10 for successful ride completion.
- +5 for reduced passenger waiting time.
- +3 for better driver utilization.
- -5 for ride cancellations.
- -3 for idle drivers.

Environmental Uncertainty

Demand changes due to traffic, weather, holidays, and special events, making the environment unpredictable.

Ethical Concerns

Dynamic pricing should remain fair. Excessive surge pricing during emergencies may create social and ethical issues, so pricing policies should balance profitability and customer fairness.

6. Smart Home Automation

States

The smart home monitors:

- Indoor temperature.
- Room occupancy.
- Outdoor weather.
- Electricity usage.
- Time of day.
- Humidity.

Actions

The system can:

- Turn lights ON/OFF.
- Increase or decrease heating.
- Start or stop air conditioning.
- Adjust fan speed.

Reward Function

- +5 for maintaining comfortable temperature.
- +5 for reducing energy consumption.
- -3 for unnecessary power usage.
- -5 for uncomfortable room conditions.

Handling Conflicting Objectives

Comfort and energy saving often conflict. A weighted reward function balances both objectives so that the system minimizes electricity consumption without reducing user comfort.

Evaluation

Reward shaping improves learning speed and helps the agent select energy-efficient actions while maintaining a comfortable indoor environment.

7. Strategy Game Agent

MDP Components

- **States:** Game board, player resources, available units, enemy positions, and current score.
- **Actions:** Attack, defend, build structures, collect resources, or upgrade units.
- **Environment:** The game world changes according to both player and opponent actions.
- **Policy:** Selects actions that maximize the probability of winning.

Reward Structure

- +100 for winning the game.
- +20 for capturing objectives.
- +10 for collecting resources.
- -20 for losing important units.
- -100 for losing the match.

Evaluation

Long-term rewards encourage strategic planning instead of immediate gains. Exploration helps discover new winning strategies, while exploitation focuses on proven successful actions.

8. Fraud Detection System

State Space

Each transaction is represented by:

- Transaction amount.
- Transaction location.
- Customer purchase history.
- Device information.
- Time of transaction.
- Merchant category.

Action Space

The RL system can:

- Approve transaction.
- Reject transaction.
- Request additional verification.

- Flag transaction for manual review.

Reward Function

- +10 for correctly detecting fraud.
- +5 for approving genuine transactions.
- -15 for missing fraudulent activity.
- -8 for incorrectly rejecting legitimate customers.

Challenges

Fraudulent transactions are very rare compared to normal transactions, creating an imbalanced dataset. Feedback may also arrive after investigations, resulting in delayed rewards.

Evaluation

False positives reduce customer satisfaction, while false negatives cause financial losses. Therefore, the RL system must carefully balance security and user convenience.

9. Personalized Rehabilitation System

States

The patient's condition includes:

- Recovery progress.
- Pain level.
- Mobility.
- Previous exercise performance.
- Age and health condition.

Actions

The RL agent recommends:

- Easy exercises.
- Moderate exercises.
- Advanced exercises.
- Rest periods.

Reward Function

- +10 for improved recovery.
- +5 for successful exercise completion.
- -5 for increased pain.
- -10 for injury or unsafe exercise.

Analysis

Patient improvement occurs gradually, so rewards are delayed. Recovery rates also vary among patients, making the environment highly personalized.

Evaluation

The system should prioritize patient safety, protect medical data privacy, and ensure that recommendations do not increase injury risk. Human medical supervision should always remain available.

10. Smart Grid Electricity Management

MDP Components

- **States:** Electricity demand, renewable energy generation, battery storage level, and grid load.
- **Actions:** Increase or decrease power supply, charge batteries, discharge batteries, or redistribute electricity.
- **Environment:** The smart grid includes homes, industries, renewable energy sources, and storage systems.
- **Policy:** Determines the best power distribution strategy for each demand condition.

Reward Function

- +10 for balancing electricity supply and demand.
- +5 for reducing peak load.
- +5 for efficient renewable energy usage.
- -10 for power outages.
- -5 for energy wastage.

Evaluation

The reward function encourages efficient electricity distribution while maintaining reliability. Large smart grids contain thousands of devices, making scalability and real-time decision-making major challenges. Efficient RL algorithms help improve grid stability, reduce operational costs, and support sustainable energy management.

